

PDF 076

$$\{U_n(x)\}$$

$$[-1, 1]$$

$$\rho(x) = \sqrt{1 - x^2}$$

$$U_0(x) = 1, \quad U_1(x) = 2x,$$

$$U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x) \quad (n = 1, 2, \dots).$$

$$[0, \infty)$$

$$\rho(x) = e^{-x}$$

$$L_n(x) = e^n \frac{d^n}{dx^n} (x^n e^{-x}). \quad (3.4.11)$$

$$\int_0^\infty e^{-x} L_n(x) L_m(x) dx = \begin{cases} 0, & m \neq n, \\ (n!)^2, & m = n \end{cases}$$

$$L_0(x) = 1, \quad L_1(x) = 1 - x,$$

$$L_{n+1}(x) = (1 + 2n - x)L_n(x) - n^2 L_{n-1}(x) \quad (n = 1, 2, \dots).$$

$$(-\infty, \infty)$$

$$\rho(x) = e^{-x^2}$$

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}). \quad (3.4.12)$$

$$\int_{-\infty}^{+\infty} e^{-x^2} H_m(x) H_n(x) dx = \begin{cases} 0, & m \neq n, \\ 2^n n! \sqrt{\pi}, & m = n, \end{cases}$$

$$H_0(x) = 1, \quad H_1(x) = 2x,$$

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \quad (n = 1, 2, \dots).$$

$$f(x) \in C[a, b]$$

$$\{g_0(x), g_1(x), \dots, g_n(x)\}$$

$$S_n(x) = a_0 g_0(x) + a_1 g_1(x) + \dots + a_n g_n(x). \quad (3.5.1)$$

$$\{g_k(x)\}$$

$$a_k = \frac{(f, g_k)}{(g_k, g_k)} \quad (k = 0, 1, \dots, n), \quad (3.5.2)$$

$$f(x)$$

$$e^n$$

$$n = 0$$

$$L_0(x) = e^{-x}$$

$$L_0(x) = 1$$

$$e^x$$

PDF 077

$$S_n(x)=\sum_{k=0}^n\frac{(f,g_k)}{(g_k,g_k)}g_k(x).$$

(3.5.3)

$$\|\delta_n\|_2=\|f-S_n\|_2=\sqrt{\|f\|_2^2-\sum_{k=0}^n\frac{(f,g_k)}{(g_k,g_k)}(f,g_k)}.$$

(3.5.4)

$f(x)$
 $[a,b]$
 $\{g_k(x)\}$
 a_k
 $k=0,1,\cdots$
 $f(x)$

$$f(x)\sim\sum_{k=0}^\infty a_k g_k(x).$$

(3.5.5)

a_k
 $f(x)\in C[a,b]$
 $S_n(x)$
 $f(x)$
 a_k
 n
 n
 a_k
 $f(x)$
 $f(x)$
 $f(x)$
 $f(x)\in C[-1,1]$
 $\{P_0(x),P_1(x),\cdots\}$
 $S_n^*(x)$

$$S_n^*(x)=a_0^*P_0(x)+a_1^*P_1(x)+\cdots+a_n^*P_n(x),$$

(3.5.6)

$$a_k^*=\frac{(f,P_k)}{(P_k,P_k)}=\frac{2k+1}{2}\int_{-1}^1f(x)P_k(x)\,\mathrm{d}x.$$

(3.5.7)

$$\|\delta_k\|_2^2=\int_{-1}^1f^2(x)\,\mathrm{d}x-\sum_{k=0}^n\frac{2}{2k+1}(a_k^*)^2.$$

(3.5.8)

$f(x)=\mathrm{e}^x$
 $[-1,1]$
 (f,P_k)
 $k=0,1,2,3$

$$(f,P_0)=\int_{-1}^1\mathrm{e}^x\,\mathrm{d}x=\mathrm{e}-\frac{1}{\mathrm{e}}\approx2.3504,$$

$$(f,P_1)=\int_{-1}^1xe^x\,\mathrm{d}x=2\mathrm{e}^{-1}\approx 0.7358,$$

$$(f,P_2)=\int_{-1}^1\left(\frac{3}{2}x^2-\frac{1}{2}\right)\mathrm{e}^x\,\mathrm{d}x=\mathrm{e}-\frac{7}{\mathrm{e}}\approx 0.1431,$$

$$(f,P_3)=\int_{-1}^1\left(\frac{5}{2}x^3-\frac{3}{2}x\right)\mathrm{e}^x\,\mathrm{d}x=37\frac{1}{\mathrm{e}}-5\mathrm{e}\approx 0.02013.$$

$$a_0^*=(f,P_0)/2=1.1752,\quad a_1^*=3(f,P_1)/2=1.1036,$$

$$a_2^*=5(f,P_2)/2=0.3578,\quad a_3^*=7(f,P_3)/2=0.07046,$$

$$\|\delta_k\|_2^2$$

$$n$$

$$\|\delta_n\|_2^2$$

PDF 078

$$S_3^*(x)=0.9963+0.9979x+0.5367x^2+0.1761x^3.$$

$$\|\delta_n\|_2=\|\mathrm{e}^x-S_3^*(x)\|_2=\sqrt{\int_{-1}^1\mathrm{e}^{2x}\,\mathrm{d}x-\sum_{k=0}^3\frac{2}{2k+1}(a_k^*)^2}\leqslant 0.0084,$$

$$\|\delta_n\|_\infty=\|\mathrm{e}^x-S_3(x)\|_\infty\leqslant 0.0112.$$

$$f(x)\in C[a,b]$$

$$f(x)$$

$$[a,b]$$

$$x=\frac{b-a}{2}t+\frac{b+a}{2}\quad (-1\leqslant t\leqslant 1),$$

$$F(t)=f\left(\frac{b-a}{2}t+\frac{b+a}{2}\right).$$

$$[-1,1]$$

$$S_n^*(t)$$

$$f(x)$$

$$[a,b]$$

$$S_n^*\left(\frac{1}{b-a}(2x-a-b)\right)$$

$$\{P_k(x)\}$$

$$[-1,1]$$

$$\{1,x,x^2,\cdots,x^k,\cdots\}$$

$$S^*(x)=a_0+a_1x+\cdots+a_nx^n$$

$$H_n$$

n
 (x_i,y_i)
 $i=0,1,\cdots,m$
 x
 y
 $y=F(x)$
 $y=F(x)$
 (x_i,y_i)
 x_i
 $\delta_i=F(x_i)-y_i$
 $i=0,1,\cdots,m$
 $\boldsymbol{\delta}=(\delta_0,\delta_1,\cdots,\delta_m)^{\text{T}}$
 $\boldsymbol{\delta}$
 $\|\boldsymbol{\delta}\|$
 $\|\boldsymbol{\delta}\|_2$
 (x_i,y_i)
 $i=0,1,\cdots,m$
 $\Phi=\text{span}\{\varphi_0,\varphi_1,\cdots,\varphi_n\}$
 $y=S^*(x)$
 $S_3(x)$
 $S_3^*(x)$

PDF 079

$$\|\boldsymbol{\delta}\|_2^2=\sum_{i=0}^m\delta_i^2=\sum_{i=0}^m[S^*(x_i)-y_i]^2=\min_{S(x)\in\varphi}\sum_{i=1}^m[S(x_i)-y_i]^2,$$

(3.6.1)

$$S(x)=a_0\varphi_0(x)+a_1\varphi_1(x)+\cdots+a_n\varphi_n(x)\quad(n<m).$$

(3.6.2)

$S(x)$
 (x_i,y_i)
 $S(x_i)$
 $S(x)$
 $\varphi_k(x)$
 k
 $S(x)$
 n
 $\|\boldsymbol{\delta}\|_2^2$

$$\|\boldsymbol{\delta}\|_2^2=\sum_{i=0}^m\omega(x_i)[S(x_i)-f(x_i)]^2.$$

(3.6.3)

$\omega(x)\geqslant 0$
 $[a,b]$
 $(x_i,f(x_i))$
 $\omega(x_i)$
 $(x_i,f(x_i))$

$$S(x)$$
$$y=S^*(x)$$

$$I(a_0,a_1,\cdots,a_n)=\sum_{i=0}^m\omega(x_i)\left[\sum_{j=0}^na_j\varphi_j(x_i)-f(x_i)\right]^2\tag{3.6.4}$$

$$(a_0^*,a_1^*,\cdots,a_n^*)$$
$$\frac{\partial I}{\partial a_k}=2\sum_{i=0}^m\omega(x_i)\left[\sum_{j=0}^na_j\varphi_j(x_i)-f(x_i)\right]\varphi_k(x_i)=0\quad(k=0,1,\cdots,n).$$
$$(\varphi_j,\varphi_k)=\sum_{i=0}^m\omega(x_i)\varphi_j(x_i)\varphi_k(x_i),\tag{3.6.5}$$

$$(f,\varphi_k)=\sum_{i=0}^m\omega(x_i)f(x_i)\varphi_k(x_i)\equiv d_k\quad(k=0,1,\cdots,n),$$
$$\sum_{j=0}^n(\varphi_k,\varphi_j)a_j=d_k\quad(k=0,1,\cdots,n).\tag{3.6.6}$$

$$G\boldsymbol{a}=\boldsymbol{d},$$

$$\boldsymbol{a}=(a_0,a_1,\cdots,a_n)^{\mathrm{T}},\quad \boldsymbol{d}=(d_0,d_1,\cdots,d_n)^{\mathrm{T}},$$

$$G=\begin{pmatrix}(\varphi_0,\varphi_0)&(\varphi_0,\varphi_1)&\cdots&(\varphi_0,\varphi_n)\\(\varphi_1,\varphi_0)&(\varphi_1,\varphi_1)&\cdots&(\varphi_1,\varphi_n)\\ \vdots&\vdots&&\vdots\\(\varphi_n,\varphi_0)&(\varphi_n,\varphi_1)&\cdots&(\varphi_n,\varphi_n)\end{pmatrix},\tag{3.6.7}$$

$$i=1$$
$$i=0$$
$$i=0$$
$$S(x)\in\varphi$$
$$\Phi$$

PDF 080

$$\varphi_0,\varphi_1,\cdots,\varphi_n$$
$$|G|\neq 0$$

$$a_k=a_k^*\quad(k=0,1,\cdots,n),$$

$$f(x)$$

$$S^*(x)=a_0^*\varphi_0(x)+a_1^*\varphi_1(x)+\cdots+a_n^*\varphi_n(x).$$

$$S^*(x)$$
$$S(x)$$

$$\sum_{i=0}^m \omega(x_i)[S^*(x_i)-f(x_i)]^2 \leq \sum_{i=0}^m \omega(x_i)^*[S(x_i)-f(x_i)]^2,$$

$$S^*(x)$$

$$x_i$$

$$f_i$$

$$\omega_i$$

$$S_1(x)=a_0+a_1x$$

$$m=4,n=1,\varphi_0(x)=1,\varphi_1(x)=x$$

$$(\varphi_0,\varphi_0)=\sum_{i=0}^4\omega_i=8,\quad (\varphi_0,\varphi_1)=(\varphi_1,\varphi_0)=\sum_{i=0}^4\omega_ix_i=22,$$

$$(\varphi_1,\varphi_1)=\sum_{i=0}^4\omega_ix_i^2=74,\quad (\varphi_0,f)=\sum_{i=0}^4\omega_if_i=47,$$

$$(\varphi_1,f)=\sum_{i=0}^4\omega_ix_if_i=145.5.$$

$$\begin{cases} 8a_0+22a_1=47,\\ 22a_0+74a_1=145.5, \end{cases}$$

$$a_0=2.77,\quad a_1=1.13.$$

$$S_1^*(x)=2.77+1.13x.$$

$$y=S_1^*(x)$$

$$x,y,0$$

$$2,4,6$$

$$2,4,6,8,10$$

$$y=S_1^*(x)$$

$$y$$

$$t$$

$$y=F(t)$$

$$t/\mathrm{min}$$

$$y/\times 10^{-3}$$

$$t\rightarrow\infty$$

$$y$$

$$\omega(x_i)$$

$$\omega(x_i)[S(x_i)-f(x_i)]^2$$

PDF 081

$$y=F(t)$$

$$t=0$$

$$y=F(t)$$

$$1/y=a+b/t$$

$y=t/(at+b)$
 a,b
 $\bar{y}=1/y,x=1/t$
 x
 $S_1(x)=a+bx$
 $(x_i,\bar{y}_i)$
 $i=1,2,\cdots,16$
 $x_i,\bar{y}_i$
 (t_i,y_i)

$$\begin{cases} 16a+3.38073b=1.8372\times10^3,\\ 3.38073a+1.58435b=0.52886\times10^3, \end{cases}$$

$$a=80.6621,\quad b=161.6822.$$

$$y=t/(80.6621t+161.6822)=F^{(1)}(t),$$

$$\delta_i^{(1)}=y_i-F^{(1)}(t_i)\quad (i=1,2,\cdots,16).$$

y
 t
 0
 $2,4,6,8,10,12,14,16,18$
 $2,4,6,8,10,12$

$$y=ae^{b/t}.$$

$t\rightarrow\infty$
 $y\rightarrow a$
 $t\rightarrow 0$
 $b<0$
 $y\rightarrow 0$
 t
 y
 a
 b
 $\ln y=\ln a+b/t$

$$\hat{y}=\ln y,\quad A=\ln a,\quad x=1/t,$$

(t_i,y_i)
 $(x_i,\hat{y}_i)$
 (x_i,y_i)
 $S_1(x)=A+bx$
 $A=-4.48072,b=-1.0567$

$$a=\mathrm{e}^A=11.3253\times10^{-3},$$

$$y=11.3253\times 10^{-3}\mathrm{e}^{-1.0567t}=F^{(2)}(t),$$

$$\delta_i^{(2)}=y_i-F^{(2)}(t_i)\quad (i=1,2,\cdots,16).$$

$$\|\boldsymbol{\delta}^{(1)}\|_\infty=\max_i|\delta_i^{(1)}|=0.568\times 10^{-3},$$

$$\|\boldsymbol{\delta}^{(2)}\|_\infty=\max_i|\delta_i^{(2)}|=0.277\times 10^{-3},$$

$$\|\boldsymbol{\delta}^{(1)}\|_2=\sqrt{\sum_{i=1}^m(\delta_i^{(1)})^2}=1.19\times 10^{-3},$$

$$\|\boldsymbol{\delta}^{(2)}\|_2=\sqrt{\sum_{i=1}^m(\delta_i^{(2)})^2}=0.34\times 10^{-3},$$

$$F^{(2)}(t)$$

$$-1.0567t$$

$$y=ae^{b/t}$$

$$b=-1.0567$$

$$\mathrm{e}^{-1.0567/t}$$

$$t\rightarrow\infty$$

$$(x_i,y_i)$$

$$(x_i,\hat{y}_i)$$

PDF 082

$$\|\boldsymbol{\delta}^{(2)}\|_2$$

$$\|\boldsymbol{\delta}^{(2)}\|_\infty$$

$$y=F^{(2)}(t)$$

$$G$$

$$\varphi_0(x),\varphi_1(x),\cdots,\varphi_n(x)$$

$$\{x_i\}$$

$$i=0,1,\cdots,m$$

$$\omega(x_i)$$

$$i=0,1,\cdots,m$$

$$(\varphi_j,\varphi_k)=\sum_{i=0}^m\omega(x_i)\varphi_j(x_i)\varphi_k(x_i)=\left\{\begin{array}{ll}0,&j\neq k,\\A_k>0,&j=k,\end{array}\right.\tag{3.6.8}$$

$$a_k^*=\frac{(f,\varphi_k)}{(\varphi_k,\varphi_k)}=\frac{\displaystyle\sum_{i=0}^m\omega(x_i)f(x_i)\varphi_k(x_i)}{\displaystyle\sum_{i=0}^m\omega(x_i)\varphi_k^2(x_i)}\quad (k=0,1,\cdots,n),\tag{3.6.9}$$

$$\|\delta\|_2^2=\|f\|_2^2-\sum_{k=0}^nA_k(a_k^*)^2.$$

$x_0,x_1,\cdots,x_m$
 $\omega(x)>0$
 $\omega(x)$
 $\{P_n(x)\}$
 $n\leqslant m$
 $P_k(x)$

$$\left\{\begin{array}{l}P_0(x)=1,\\P_1(x)=(x-\alpha_1)P_0(x),\\P_{k+1}(x)=(x-\alpha_{k+1})P_k(x)-\beta_kP_{k-1}(x)\quad(k=1,2,\cdots,n-1).\end{array}\right.\tag{3.6.10}$$

$P_k(x)$
 1
 k
 $P_k(x)$

$$\left\{\begin{array}{l}a_{k+1}=\frac{\sum\limits_{i=0}^m\omega(x_i)x_iP_k^2(x_i)}{\sum\limits_{i=0}^m\omega(x_i)P_k^2(x_i)}=\frac{(xP_k(x),P_k(x))}{(P_k(x),P_k(x))}=\frac{(xP_k,P_k)}{(P_k,P_k)},\\[10pt]\beta_k=\frac{\sum\limits_{i=0}^m\omega(x_i)P_k^2(x_i)}{\sum\limits_{i=0}^m\omega(x_i)P_{k-1}^2(x_i)}=\frac{(P_k,P_k)}{(P_{k-1},P_{k-1})}\end{array}\right.\quad(k=0,1,\cdots,n-1).\tag{3.6.11}$$

a_{k+1}
 α_{k+1}
 $k=0,1,\cdots,n-1$
 β_k
 $k=0$
 P_{-1}
 $\beta_1,\ldots,\beta_{n-1}$

PDF 083

$\{P_k(x)\}$
 a_1

$$\begin{aligned}(P_l,P_1)&=(P_0,xP_0)-\alpha_1(P_0,P_0)\\&=(P_0,xP_0)-\frac{(xP_0,P_0)}{(P_0,P_0)}(P_0,P_0)=0.\end{aligned}$$

$(P_1,P_s)=0$
 $l\neq s$
 $s=0,1,\cdots,l-1$
 $l=0,1,\cdots,k$
 $k< n$

$$(P_{k+1}, P_s) = 0$$

$$s = 0, 1, \dots, k$$

$$(P_{k+1}, P_s) = ((x - \alpha_{k+1})P_k, P_s) - \beta_k(P_{k-1}, P_s)$$

$$= (xP_k, P_s) - \alpha_{k+1}(P_k, P_s) - \beta_k(P_{k-1}, P_s), \quad (3.6.12)$$

$$0 \leq s \leq k-2$$

$$(P_k, P_s) = 0, \quad (P_{k-1}, P_s) = 0.$$

$$xP_s(x)$$

$$1$$

$$s+1$$

$$P_0, P_1, \dots, P_{s+1}$$

$$s+1 \leq k-1$$

$$(xP_k, P_s) \equiv (P_k, xP_s) = 0,$$

$$s \leq k-2$$

$$(P_{k+1}, P_s) = 0$$

$$(P_{k+1}, P_{k-1}) = (xP_k, P_{k-1}) - \alpha_{k+1}(P_k, P_{k-1}) - \beta_k(P_{k-1}, P_{k-1}), \quad (3.6.13)$$

$$(P_k, P_{k-1}) = 0,$$

$$(xP_k, P_{k-1}) = (P_k, xP_{k-1}) = \left(P_k, P_k + \sum_{j=0}^{k-1} C_j P_j \right) = (P_k, P_k).$$

$$\beta_k$$

$$(P_{k+1}, P_{k-1}) = (xP_k, P_{k-1}) - \beta_k(P_{k-1}, P_{k-1}) = (P_k, P_k) - (P_k, P_k) = 0.$$

$$(P_{k+1}, P_k) = (xP_k, P_k) - \alpha_{k+1}(P_k, P_k) - \beta_k(P_k, P_{k-1})$$

$$= (xP_k, P_k) - \frac{(xP_k, P_k)}{(P_k, P_k)}(P_k, P_k) = 0.$$

$$\{P_k(x)\}$$

$$k = 0, 1, \dots, n; n \leq m$$

$$\{x_i\}$$

$$\{P_k(x)\}$$

$$P_k(x)$$

$$a_k = \frac{(f, P_k)}{(P_k, P_k)} = \frac{\sum_{i=0}^m \omega(x_i) f(x_i) P_k(x_i)}{\sum_{i=0}^m \omega(x_i) P_k^2(x_i)} \quad (k = 0, 1, \dots, n),$$

$$\alpha_k^* P_k(x)$$

$$F(x)$$

$$y=F(x)=\alpha_0^*P_0(x)+\alpha_1^*P_1(x)+\cdots+\alpha_n^*P_n(x),$$

$$(P_l,P_1)$$

$$l$$

$$(P_0,P_1)$$

$$(P_1,P_s)=0$$

$$l\neq s$$

$$(P_l,P_s)=0$$

$$a_k$$

$$\alpha_k^*$$

$$\alpha_k$$

PDF 084

$$n$$

$$1$$

$$y=f(x_1,x_2,\cdots,x_l)$$

$$(x_{1i},x_{2i},\cdots,x_{li},y_i)$$

$$i=1,2,\cdots,m$$

$$\omega_i>0$$

$$i=1,2,\cdots,m$$

$$S_n(x_1,x_2,\cdots,x_l)=\sum_{k=1}^na_k\varphi_k(x_1,x_2,\cdots,x_l),\quad n\leqslant m,$$

$$F(a_0,a_1,\cdots,a_n)=\sum_{i=1}^m\omega_i[y_i-S_n(x_{1i},x_{2i},\cdots,x_{li})]^2$$

$$a_0,a_2,\cdots,a_n$$

$$(\varphi_k,\varphi_j)=\sum_{i=1}^m\omega_i\varphi_k(x_{1i},x_{2i},\cdots,x_{li})\varphi_i(x_{1i},x_{2i},\cdots,x_{li}).$$

$$a_k$$

$$k=0,1,\cdots,n$$

$$S_n(x_1,x_2,\cdots,x_l)$$

$$S_n(x_1,x_2,\cdots,x_l)$$

$$f(x_1,x_2,\cdots,x_l)$$

$$f(x)$$

$$f(x)$$

$$2\pi$$

$$S_n(x)=\frac{1}{2}a_0+a_1\cos x+b_1\sin x+\cdots+a_n\cos nx+b_n\sin nx\tag{3.7.1}$$

$$1,\cos x,\sin x,\cdots,\cos kx,\sin kx,\cdots$$

$$[0,2\pi]$$

$$f(x)$$

$$[0,2\pi]$$

$$S_n(x)$$

$$S_n$$

$$k=1$$

$$F(a_0,a_1,\ldots,a_n)$$

$$a_k\;(k=0,\ldots,n)$$

$$a_0,a_2,\ldots,a_n$$

$$i$$

$$j$$

$$\varphi_j$$

PDF 085

$$\left\{\begin{array}{l}a_k=\frac{1}{\pi}\int_0^{2\pi}f(x)\cos kx\,\mathrm{d}x\quad(k=0,1,\cdots,n),\\b_k=\frac{1}{\pi}\int_0^{2\pi}f(x)\sin kx\,\mathrm{d}x\quad(k=1,2,\cdots,n),\end{array}\right.\tag{3.7.2}$$

$$a_k,b_k$$

$$f(x)$$

$$\frac{1}{2}a_0+\sum_{k=1}^{\infty}(a_k\cos kx+b_k\sin kx)\tag{3.7.3}$$

$$f'(x)$$

$$[0,2\pi]$$

$$f(x)$$

$$f(x)$$

$$\left\{x_j=\frac{2\pi}{N}j,\;j=0,1,\cdots,N-1\right\}$$

$$x_j=\frac{2\pi j}{2m+1}\quad(j=0,1,\cdots,2m),$$

$$0\leqslant k,l\leqslant m$$

$$\left\{\begin{array}{l}\sum\limits_{j=0}^{2m}\sin lx_j\sin kx_j=\left\{\begin{array}{ll}0,&l\neq k,\;l=k=0,\\ \frac{2m+1}{2},&l=k\neq 0,\end{array}\right.\\ \sum\limits_{j=0}^{2m}\cos lx_j\cos kx_j=\left\{\begin{array}{ll}0,&l\neq k,\\ \frac{2m+1}{2},&l=k\neq 0,\\ 2m+1,&l=k=0,\end{array}\right.\\ \sum\limits_{j=0}^{2m}\cos lx_j\sin kx_j=0,\quad 0\leqslant k,j\leqslant m.\end{array}\right.$$

$$\{1,\cos x,\sin x,\cdots,\cos mx,\sin mx\}$$

$$\left\{x_j=\frac{2\pi j}{2m+1}\right\}$$

$$f_j=f(x_j)$$

$$j=0,1,\cdots,2m$$

$$f(x)$$

$$S_n(x)=\frac{1}{2}a_0+\sum_{k=1}^n(a_k\cos kx+b_k\sin kx),\quad n< m,$$

$$\begin{cases} a_k=\frac{2}{2m+1}\sum_{j=0}^{2m}f_j\cos\frac{2\pi jk}{2m+1} & (k=0,1,\cdots,n),\\ b_k=\frac{2}{2m+1}\sum_{j=0}^{2m}f_j\sin\frac{2\pi jk}{2m+1} & (k=1,\cdots,n). \end{cases}\tag{3.7.4}$$

$$n=m$$

$$S_m(x_j)=f_j$$

$$j=0,1,\cdots,2m$$

$$S_m(x)=\frac{1}{2}a_0+\sum_{k=1}^m(a_k\cos kx+b_k\sin kx)$$

$$0\leqslant k,j\leqslant m$$

$$j$$

$$k,l$$

$$0\leqslant k,l\leqslant m$$

PDF 086

$$f(x)$$

$$2\pi$$

$$N$$

$$x_j=\frac{2\pi}{N}j$$

$$j=0,1,\cdots,N-1$$

$$f_j=f\left(\frac{2\pi}{N}j\right)$$

$$\mathrm{e}^{\mathrm{i}jx}=\cos(jx)+\mathrm{i}\sin(jx)\quad(j=0,1,\cdots,N-1;\,\mathrm{i}=\sqrt{-1}),$$

$$\{1,\mathrm{e}^{\mathrm{i}x},\cdots,\mathrm{e}^{\mathrm{i}(N-1)x}\}$$

$$[0,2\pi]$$

$$\mathrm{e}^{\mathrm{i}jx}$$

$$x_k=\frac{2\pi}{N}k$$

$$k=0,1,\cdots,N-1$$

$$\mathrm{e}^{\mathrm{i}jx_k}$$

$$\boldsymbol{\varphi}_j=\left(1,\mathrm{e}^{\mathrm{i}j\frac{2\pi}{N}},\cdots,\mathrm{e}^{\mathrm{i}j\frac{2\pi}{N}(N-1)}\right)^{\mathrm{T}}.$$

$$j=0,1,\cdots,N-1$$

$$N$$

$$\boldsymbol{\varphi}_0,\boldsymbol{\varphi}_1,\cdots,\boldsymbol{\varphi}_{N-1}$$

$$(\boldsymbol{\varphi}_l,\boldsymbol{\varphi}_s)=\sum_{k=0}^{N-1}\mathrm{e}^{\mathrm{i}l\frac{2\pi}{N}k}\mathrm{e}^{-\mathrm{i}s\frac{2\pi}{N}k}=\sum_{k=0}^{N-1}\mathrm{e}^{\mathrm{i}(l-s)\frac{2\pi}{N}k}=\begin{cases}0,&l\neq s,\\N,&l=s.\end{cases}\tag{3.7.5}$$

$$r=\mathrm{e}^{\mathrm{i}(l-s)\frac{2\pi}{N}}$$

$0 \leqslant l,s \leqslant N-1$

$0 \leqslant l \leqslant N-1, \quad -(N-1) \leqslant -s \leqslant 0,$

$-(N-1) \leqslant l-s \leqslant N-1, \quad -1 < -\frac{N-1}{N} \leqslant \frac{l-s}{N} \leqslant \frac{N-1}{N} < 1.$

$l-s \neq 0$
 $r \neq 1$
 $r^N = \mathrm{e}^{\mathrm{i}(l-s)2\pi} = 1$

$(\varphi_l,\varphi_s)=\sum_{k=0}^{N-1}r^k=\frac{1-r^N}{1-r}=0.$

$l=s$
 $r=1$
 $(\varphi_s,\varphi_s)=\sum_{k=0}^{N-1}r^k=N$
 $\varphi_0,\varphi_1,\cdots,\varphi_{N-1}$
 $f(x)$
 $\left\{x_j=\frac{2\pi}{N}j,\,j=0,1,\cdots,N-1\right\}$

$$S(x)=\sum_{k=0}^{n-1}C_k\mathrm{e}^{\mathrm{i}kx}, \quad n\leqslant N, \tag{3.7.6}$$

$$C_k=\frac{1}{N}\sum_{j=0}^{N-1}f_j\mathrm{e}^{-\mathrm{i}kj\frac{2\pi}{N}} \quad (k=0,1,\cdots,N-1). \tag{3.7.7}$$

$n=N$
 $S(x)$
 $f(x)$
 x_j
 $j=0,1,\cdots,N-1$
 $S(x_j)=f(x_j)$

$$f_j=\sum_{k=0}^{N-1}C_k\mathrm{e}^{\mathrm{i}k\frac{2\pi}{N}j} \quad (j=0,1,\cdots,N-1). \tag{3.7.8}$$

$\{f_j\}$
 $\{C_k\}$
 $f(x)$

PDF 087

$\{C_k\}$
 $\{f_j\}$
 $\{f_k\}$
 $\{C_j\}$
 $\{C_j\}$
 $\{f_k\}$

a_k, b_k

$$C_j = \sum_{k=0}^{N-1} x_k w^{kj} \quad (j = 0, 1, \cdots, N-1), \tag{3.7.9}$$

$w = \exp(-\mathrm{i}2\pi/N)$

$w = \exp(\mathrm{i}2\pi/N)$

$\{x_k\}$

$k = 0, 1, \cdots, N-1$

C_j

N

N

N

C_j

N^2

N

$ab+ac=a(b+c)$

C_j

N^2

$\exp(\mathrm{i}2\pi kj/N), \ j, k = 0, 1, \cdots, N-1$

N

$w^0, w^1, \cdots, w^{N-1}$

$N=2^p$

$N/2$

w^r

x_k

w^r

m

N

q

r

$m=qN+r$

r

m

N

$w = \exp(\mathrm{i}2\pi/N), w^N = \mathrm{e}^{\mathrm{i}2\pi} = 1$

$w^m = (w^N)^qw^r = w^r$

w^m

w

N

r

m

$N=2^3$

$0 \leqslant k, j \leqslant N-1 = 2^3-1 = 7$

$$C_j = \sum_{k=0}^7 x_k w^{jk} \quad (j = 0, 1, \cdots, 7). \tag{3.7.10}$$

$$k,j$$

$$k=k_22^2+k_12+k_02^0=(k_2k_1k_0),$$

$$j=j_22^2+j_12^1+j_02^0=(j_2j_1j_0),$$

$$k_r,j_r$$

$$r=0,1,2$$

$$0$$

$$1$$

$$6=2^2+2^1+0\times 2^0=(110)$$

$$k,j$$

$$C_j=C(j_2j_1j_0),x_k=x(k_2k_1k_0)$$

$$\begin{aligned} C(j_2j_1j_0) &= \sum_{k_0=0}^1 \sum_{k_1=0}^1 \sum_{k_2=0}^1 x(k_2k_1k_0)w^{(k_2k_1k_0)(j_22^2+j_12^1+j_02^0)} \\ &= \sum_{k_0=0}^1 \left\{ \sum_{k_1=0}^1 \left[\sum_{k_2=0}^1 x(k_2k_1k_0)w^{j_0(k_2k_1k_0)} \right] w^{j_1(k_1k_00)} \right\} \left(w^{j_2(k_000)} \right). \end{aligned} \tag{3.7.11}$$

$$N=2^p$$

$$N/2$$

$$\exp(\mathrm{i}2\pi kj/N)$$

$$j=1$$

$$N$$

$$w^{r+N/2}=-w^r$$

$$N/2$$

$$w$$

$$N$$

$$r$$

$$m$$

$$m$$

PDF 088

$$\left\{ \begin{aligned} A_0(k_2k_1k_0) &= x(k_2k_1k_0), \\ A_1(k_1k_0j_0) &= \sum_{k_2=0}^1 A_0(k_2k_1k_0)w^{j_0(k_2k_1k_0)}, \\ A_2(k_0j_1j_0) &= \sum_{k_1=0}^1 A_1(k_1k_0j_0)w^{j_1(k_1k_00)}, \\ A_3(j_2j_1j_0) &= \sum_{k_0=0}^1 A_2(k_0j_1j_0)w^{j_2(k_000)}, \end{aligned} \right. \tag{3.7.12}$$

$$C(j_2j_1j_0)=A_3(j_2j_1j_0).$$

$$N$$

$$C_j$$

p A_q 2 C_j $2p$ C_j $2pN$

$$w^{j_0 2^{p-1}} = w^{j_0 N/2} = (-1)^j$$

$$\begin{aligned} A_1(k_1 k_0 j_0) &= \sum_{k_2=0}^1 A_0(k_2 k_1 k_0) w^{j_0(k_2 k_1 k_0)} \\ &= A_0(0k_1 k_0) w^{j_0(0k_1 k_0)} + A_0(1k_1 k_0) w^{j_0 2^2} w^{j_0(0k_1 k_0)} \\ &= [A_0(0k_1 k_0) + (-1)^{j_0} A_0(1k_1 k_0)] w^{j_0(0k_1 k_0)}, \end{aligned}$$

$$A_1(k_1 k_0 0) = A_0(0k_1 k_0) + A_0(1k_1 k_0),$$

$$A_1(k_1 k_0 1) = [A_0(0k_1 k_0) - A_0(1k_1 k_0)] w^{(0k_1 k_0)}.$$

$$k = (0k_1 k_0) = k_1 2^1 + k_0 2^0$$

$$k = 0, 1, 2, 3$$

$$\begin{cases} A_1(2k) = A_0(k) + A_0(k + 2^2), \\ A_1(2k + 1) = [A_0(k) - A_0(k + 2^2)] w^k \end{cases} \quad (k = 0, 1, 2, 3). \quad (3.7.13)$$

 A_2

$$A_2(k_0 j_1 j_0) = [A_1(0k_0 j_0) + (-1)^{j_1} A_1(1k_0 j_0)] w^{j_1(0k_0 0)},$$

$$A_2(k_0 0 j_0) = A_1(0k_0 j_0) + A_1(1k_0 j_0),$$

$$A_2(k_0 1 j_0) = [A_1(0k_0 j_0) - A_1(1k_0 j_0)] w^{(0k_0 0)}.$$

$$\begin{cases} A_2(k 2^2 + j) = A_1(2k + j) + A_1(2k + j + 2^2), & k = 0, 1, \\ A_2(k 2^2 + j + 2) = [A_1(2k + j) - A_1(2k + j + 2^2)] w^{2k}, & j = 0, 1. \end{cases} \quad (3.7.14)$$

 A_3

$$A_3(j_2 j_1 j_0) = A_2(0j_1 j_0) + (-1)^{j_2} A_2(1j_1 j_0),$$

$$A_3(0j_1 j_0) = A_2(0j_1 j_0) + A_2(1j_1 j_0),$$

$$A_3(1j_1 j_0) = A_2(0j_1 j_0) - A_2(1j_1 j_0).$$

PDF 089

$$\begin{cases} A_3(j) = A_2(j) + A_2(j + 2^2), \\ A_3(j + 2^2) = A_2(j) - A_2(j + 2^2) \end{cases} \quad (j = 0, 1, 2, 3). \quad (3.7.15)$$

$$A_0(k) = x(k) = x_k$$

$$k = 0, 1, \dots, 7$$

$$A_3(j) = C_j$$

$$j = 0, 1, \dots, 7$$

$$w^0 = 1$$

$$w^1$$

$$w^2$$

$$w^3$$

$$x_k = A_0(k)$$

$$A_0(0)$$

$$A_0(1)$$

$$A_0(2)$$

$$A_0(3)$$

$$A_0(4)$$

$$A_0(5)$$

$$A_0(6)$$

$$A_0(7)$$

$$A_1$$

$$A_0(0) + A_0(4)$$

$$[A_0(0) - A_0(4)]w^0$$

$$A_0(1) + A_0(5)$$

$$[A_0(1) - A_0(5)]w^1$$

$$A_0(2) + A_0(6)$$

$$[A_0(2) - A_0(6)]w^2$$

$$A_0(3) + A_0(7)$$

$$[A_0(3) - A_0(7)]w^3$$

$$A_2$$

$$A_1(0) + A_1(4)$$

$$A_1(0) + A_1(5)$$

$$[A_1(0) - A_1(4)]w^0$$

$$[A_1(1) - A_1(5)]w^0$$

$$A_1(2) + A_1(6)$$

$$A_1(3) + A_1(7)$$

$$[A_1(2) - A_1(6)]w^2$$

$$[A_1(3) - A_1(7)]w^2$$

$$C_i = A_3(j)$$

$$(0) + (4)$$

$$(1) + (5)$$

$$(2) + (6)$$

$$(3) + (7)$$

$$(0) - (4)$$

$$(1) - (5)$$

$$(2) - (6)$$

$$(3) - (7)$$

$$N = 2^3$$

$$N = 2^p$$

$$\begin{cases} A_q(k2^q + j) = A_{q-1}(k2^{q-1} + j) + A_{q-1}(k2^{q-1} + j + 2^{p-1}), \\ A_q(k2^q + j + 2^{q-1}) = [A_{q-1}(k2^{q-1} + j) - A_{q-1}(k2^{q-1} + j + 2^{p-1})]w^{k2^{q-1}}, \end{cases} \quad (3.7.16)$$

$$q = 1, 2, \dots, p; \ k = 0, 1, \dots, 2^{p-q} - 1; \ j = 0, 1, \dots, 2^{q-1} - 1$$

$$A_q$$

$$A_q$$

$$N$$

$$A_0(m)$$

$$m = 0, 1, \dots, N - 1$$

$$q$$

$$1$$

$$p$$

$$A_p(j) = C_j$$

$$j = 0, 1, \dots, N - 1$$

$$A_q$$

$$A_p(j)$$

$$q$$

$$1$$

$$p$$

$$k$$

$$0$$

$$2^{p-q} - 1$$

$$j$$

$$0$$

$$2^{q-1} - 1$$

$$A_q$$

$$2^{p-q}2^{q/1} = N/2$$

$$A_p$$

$$w^{k2^{p-1}} = (w^{N/2})^k = (-1)^k = (-1)^0 = 1$$

$$q = p$$

$$2^{p-q} - 1 = 0$$

$$k = 0$$

$$(p-1)N/2$$

$$N^2$$

$$N : (p-1)/2$$

$$N = 2^{10}$$

$$1024 : 4.5 \approx 230 : 1$$

$$pN$$

$$A_2$$

$$\begin{array}{l} A_1(0)+A_1(5) \\ k=0,j=1 \\ A_1(1)+A_1(5) \\ C_i=A_3(j) \\ C_j \\ 2^{p-q}2^{q/1}=N/2 \\ 2^{p-q}2^{q-1}=N/2 \end{array}$$

PDF 090

$$\begin{array}{l} A_1(N),A_2(N) \\ w(N/2) \\ \{x_k\} \\ A_1(k) \\ k \\ 0 \\ N-1 \\ w^m=\exp\left(-\mathrm{i}\frac{2\pi}{N}m\right) \\ w^m=\exp\left(\mathrm{i}\frac{2\pi}{N}m\right) \\ w(m) \\ m \\ 0 \\ (N/2)-1 \\ q \\ 1 \\ p \\ q \\ k \\ 0 \\ 2^{p-q}-1 \\ j \\ 0 \\ 2^{q-1}-1 \end{array}$$

$$A_2(k2^q+j)=A_1(k2^{q-1}+j)+A_1(k2^{q-1}+j+2^{p-1}),$$

$$A_2(k2^q+j+2^{q-1})=[A_1(k2^{q-1}+j)-A_1(k2^{q-1}+j+2^{q-1})]w(k2^{q-1}),$$

$$\begin{array}{l} k \\ 0 \\ 2^{p-q}-1 \\ j \\ 0 \\ 2^{q-1} \end{array}$$

$$A_1(k2^q+j)=A_2(k2^{q-1}+j)+A_2(k2^{q-1}+j+2^{p-1}),$$

$$A_1(k2^q+j+2^{q-1})=[A_2(k2^{q-1}+j)-A_2(k2^{q-1}+j+2^{q-1})]w(k2^{q-1}),$$

$$k,j$$

$$q=p$$

$$q+1\rightarrow q$$

$$q$$

$$p=\text{偶数}$$

$$A_1(j)\rightarrow A_2(j)$$

$$C_j=A_2(j)$$

$$j=0,1,\cdots,N-1$$

$$[a,b]$$

$$2^{q-1}$$

$$2^{p-1}$$

$$j$$

$$2^{q-1}$$

$$-1$$

$$j$$

$$q=p$$

$$N$$

PDF 091

$$f(x)=\sin x$$

$$[0,\pi/2]$$

$$m\leqslant f(x)\leqslant M$$

$$m\leqslant B_n(f,x)\leqslant M$$

$$f(x)=x$$

$$B_n(f,x)=x$$

$$6$$

$$f(x)=\sin 4x$$

$$[0,2\pi]$$

$$f(x)$$

$$[a,b]$$

$$f(x)$$

$$a$$

$$\max_{0\leqslant x\leqslant 1}|x^3-ax|$$

$$f(x)=\sin x$$

$$[0,\pi/2]$$

$$f(x)=\mathrm{e}^x$$

$$[0,1]$$

$$r$$

$$p(x)=x^2+r$$

$$[-1,1]$$

$$r$$

$$f(x)=x^4+3x^3-1$$

$$[0,1]$$

$$T_n(x)=T_n(2x-1), x\in [0,1]$$

$$T_0^*(x), T_1^*(x), T_2^*(x), T_3^*(x)$$

$$\{T_n^*(x)\}$$

$$[0,1]$$

$$\rho=\frac{1}{\sqrt{x-x^2}}$$

$$f(x)$$

$$[-a,a]$$

$$n$$

$$f(x)$$

$$F_n^*(x)\in H_n$$

$$a,b$$

$$\int_0^{\frac{\pi}{2}}[ax+b-\sin x]^2\,\mathrm{d}x$$

$$f(x),g(x)\in C^1[a,b]$$

$$(f,g)=\int_a^bf'(x)g'(x)\,\mathrm{d}x$$

$$(f,g)=\int_a^bf'(x)g'(x)\,\mathrm{d}x+f(a)g(a)$$

$$\int_0^1\frac{x^6}{1+x}\,\mathrm{d}x$$

$$\int_{-1}^1 (x-ax^2)^2\,\mathrm{d}x$$

$$\int_{-1}^1|x-ax^2|\,\mathrm{d}x$$

$$\varphi_1=\mathrm{span}\{1,x\}, \varphi_2=\mathrm{span}\{x^{100},x^{101}\}$$

$$\varphi_1,\varphi_2$$

$$x^2\in C[0,1]$$

$$f(x)=|x|$$

$$[-1,1]$$

$$\varphi_1=\mathrm{span}\{1,x^2,x^4\}$$

$$u_n(x)=\frac{\sin[(n+1)\arccos x]}{\sqrt{1-x^2}}$$

$$u_{n+1}(x)=2xu_n(x)-u_{n-1}(x).$$

$$f(x)=\sin\frac{1}{2}x$$

$$[-1,1]$$

$$f(x)=\arccos x$$

$$[-1,1]$$

$$y=a+bx^2$$

$$T_n(x)$$

$$T_0^*, T_1^*, T_2^*, T_3^*$$

$$T_n^*$$

$$T_n^*(x)=T_n(2x-1)$$

PDF 092

x_i
 y_i
 t
 s
 t/s
 s/m
 y
 t
 $y = F(t)$
 t/min
 $y/ \times 10^{-4}$
 t/min
 $y/ \times 10^{-4}$
 $\{x_k\} = (4, 3, 2, 1, 0, 1, 2, 3)$
 $\{x_k\}$
 $\{C_k\}$
 $k = 0, 1, \cdots, 7$